

Rajveer Deka

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EDUCATION

The University of Texas at Austin, *BS in Electrical and Computer Engineering; Austin, TX*

Expected May 2028

- Cockrell Engineering Honors, University Honors
- **Courses:** Embedded Systems, Software Design and Implementation, Digital Logic Design, Circuit Theory*, Linear Systems and Signals*
- **GPA:** 3.932/4.000

EXPERIENCE

Tokyo Tech, *Study Abroad; Tokyo, Japan*

May 2026 – August 2026

- Studied Digital Logic Design and Probability while networking with tech industry professionals.
- Presented a final digital design project as part of an academic showcase hosted at J.F. Oberlin University in Shinjuku.

Schneider Structural Engineering, *Engineering Research Intern; Prosper, TX*

January 2025 - August 2025

- Modeled **20+ structural systems** using **AutoCAD** and **Revit** across **4 commercial/residential projects**
- Interpreted engineering data to refine design methods and ensure code compliance as part of a team of 6-8 engineers

Aakha Biologics, *Lab Assistant; Frisco, TX*

June 2024 - July 2024

- Co-developed a searchable database to categorize **100+ antibodies and antigens**, improving lab data retrieval speed by **40%**
- Operated advanced lab equipment and collaborated with a team of 5 assistants under the guidance of the senior scientist

PROJECTS

Tokyo Advertisement Sign: Scrolling LED Display

Summer 2026

- Designed and implemented a multi-mode scrolling LED sign on a Basys3 FPGA utilizing **Verilog**
- Formalized specifications into an HLSM and engineered a strict datapath and controller FSM to prevent state explosion.
- Integrated clock dividers, an edge detector, a display multiplexer, and a custom letter-to-7-segment decoder among others

Monte Carlo Estimation of π

Summer 2026

- Developed a hardware-based Monte Carlo simulation using Linear Feedback Shift Registers (LFSRs) in **Verilog** to reliably generate pseudo-random coordinate points.
- Extracted hardware simulation results across multiple samples to a .csv format and processed the final mathematical Pi estimation using Microsoft **Excel**.

Calculator Design: RCA and CLA Adders

Summer 2026

- Implemented and compared 4-bit Ripple Carry (RCA) and Carry Lookahead (CLA) adders with a clocked 5-bit output register
- Derived per-bit propagate and generate equations for the CLA and cascaded four full adders for the RCA
- Analyzed synthesized schematics in **Vivado** to quantitatively compare critical-path delay and total gate area

Hardware Blackjack Simulator & Custom PCB

Spring 2026

- Engineered a full-stack Blackjack simulator in **C++** running on an MSPM0 microcontroller, integrated onto a custom PCB.
- Interfaced the microcontroller with external hardware inputs for player actions and an LCD screen for real-time visual output of the game state.

SKILLS

Verilog, C/C++, Python, KiCAD, AutoCAD, Revit, Excel, SQL, Vivado, Embedded C, ARM Assembly, UART, SPI